

# Zihua Liu

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## PROFESSIONAL SUMMARY

Machine Learning Engineer and computer vision researcher with a Ph.D. and experience building production-grade AI systems from multimodal data processing and model training to evaluation, inference, and cloud deployment. Currently developing the end-to-end AI ecosystem for Woven by Toyota's Digital Genba, delivering scalable vision and multimodal solutions for Toyota teams and overseas customers.

## EDUCATION

### Institute of Science Tokyo (Tokyo Institute of Technology)

- Ph.D. in Engineering, Computer Vision and Deep Learning
- M.Eng. in Engineering

Tokyo, Japan

Mar 2023 – Mar 2026

Apr 2021 – Mar 2023

### South China University of Technology

- B.Eng. in Engineering

Guangzhou, China

Sep 2016 – Jun 2020

## RESEARCH & ENGINEERING EXPERIENCE

### Woven by Toyota, Inc.

- Machine Learning Engineer — Digital Genba

Tokyo, Japan

Aug 2025 – Present

- Architected an end-to-end AI ecosystem spanning multimodal data ingestion, video/frame processing, annotation and versioned datasets, model training, evaluation, inference, visualization, and deployment.
- Built production Metaflow workflows on AWS Batch and Step Functions with Dockerized GPU jobs, resumable execution, reproducible configuration, and S3/Lance artifact publication.
- Developed detection, segmentation, behavior-understanding, ReID, and graph-based multi-camera tracking capabilities using PyTorch, RF-DETR/YOLOX, Grounded SAM 2, VLMs/vLLM, and 3D vision.
- Delivered reusable Digital Genba solutions for Toyota teams and overseas customers, translating logistics and industrial-site requirements into scalable AI platform components.

### Sony Semiconductor Solutions

- R&D Research Intern

Tokyo, Japan

Jan 2025 – Mar 2025

- Built a synchronized LiDAR–image–3D box–text data pipeline and trained a multi-view diffusion model, enabling scalable street-view and BEV data generation for autonomous-perception development.

### Preferred Networks

- R&D Research Intern

Tokyo, Japan

Aug 2024 – Oct 2024

- Developed a controllable diffusion-based background generation method and benchmarked it against established baselines, validating robust generalization across datasets and diverse artistic foreground styles.

### SenseTime Japan

- Computer Vision Research Intern

Tokyo, Japan

Jun 2023 – Nov 2023

- Co-developed VSRD to train monocular 3D object detectors from 2D supervision instead of costly 3D labels, using multi-view pseudo-labeling and volumetric silhouette rendering; published at CVPR 2024.

### Megvii Research

- Computer Vision Research Intern

Shanghai, China

Feb 2022 – Apr 2022

- Increased parking-slot detection recall by 10% through geometric constraints and targeted augmentation, improving robustness near image boundaries and under occlusion; adopted in Megvii's automatic-parking perception pipeline.

## SELECTED PUBLICATIONS

- **Zihua Liu** and Masatoshi Okutomi, "StereoSplat+: Feed-Forward Stereo Gaussian Splatting with Diffusion-Assisted Progressive Inference," *IRIS*, 2026.
- **Zihua Liu** et al., "DMS: Diffusion-Based Multi-Baseline Stereo Generation for Improving Self-Supervised Depth Estimation," *ICCV*, 2025.
- **Zihua Liu** et al., "VSRD: Volumetric Silhouette Rendering for Weakly Supervised 3D Object Detection," *CVPR*, 2024.
- **Zihua Liu** et al., "CFDNet: A Generalizable Foggy Stereo Matching Network with Contrastive Feature Distillation," *ICRA*, 2024.

## HONORS & AWARDS

- Best Paper Award, Symposium on Sensing via Image Information (SSII), 2026; Best Student Paper Award, International Conference on Multimedia Modeling (MMM), 2025
- Tokyo Tech Advanced Human Resource Development Fellowship, 2023–2026; SCIE Student Award, 2023

## TECHNICAL SKILLS

- **Programming & ML:** Python, C++, CUDA, PyTorch, distributed training, diffusion models, VLMs, object detection, segmentation, ReID, multi-camera tracking, NeRF, 3D Gaussian Splatting
- **Data & Production:** Metaflow, AWS Batch, Step Functions, S3, Lance, Docker, vLLM, Git, Linux, CI/CD, model evaluation and deployment
- **Languages:** English, Japanese, Chinese